

Real-Time Log Streaming Pipeline on AWS

Project Goal

The primary objective of this project was to establish a real-time log streaming pipeline on AWS. The pipeline is designed to continuously monitor and stream application logs generated on an EC2 instance to an S3 bucket for storage and potential analysis. To achieve this, the AWS Kinesis Agent is installed on the EC2 instance to tail and forward logs to a Kinesis Data Stream. This stream then feeds into a Kinesis Data Firehose delivery stream, which buffers and delivers the logs to S3. The resulting setup provides a scalable, low-latency solution for ingesting logs from servers into object storage, supporting use cases such as monitoring, auditing, and analytics in a cloud environment.

Steps Performed

1. S3 Bucket Creation

An S3 bucket named logbucket-198429 was created in the US East (N. Virginia) region (us-east-1) on December 27, 2025, at 4:19:44 PM UTC. This bucket serves as the final destination for streamed logs.

	logbucket-198429	US East (N. Virginia) us-east-1	December 27, 2025, 16:19:44 (UTC-06:00)
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2. Kinesis Data Stream Provisioning

A Kinesis Data Stream named logstream was set up with a capacity of 1024 KiB (1 MiB), activated, and configured with a one-day retention period. This stream acts as the initial ingestion point for log data from the EC2 instance.

	logstream	 Active	Provisioned	1024 KiB	-	1	No	1 day
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3. EC2 Instance Access Preparation

To facilitate access to the EC2 instance, the SSH key file `kinesis-key.pem` was secured using `chmod 400 kinesis-key.pem`, ensuring read-only owner permissions. An SSH connection was then established to the EC2 instance at IP address `98.84.xx.xx` using the command `ssh -i kinesis-key.pem ec2-user@98.84.xx.xx`. Successful login was confirmed by access to the instance's prompt.

```
chmod 400 kinesis-key.pem
ssh -i kinesis-key.pem ec2-user@98.84.186.133
ec2-user@98.84.186.133:~$
```

4. IAM Role Creation for Firehose

During the setup of Firehose in the AWS console, an IAM role named `IAMRoleKinesis` was selected or created. This role grants Firehose the necessary permissions to access the Kinesis Data Stream and write to the S3 bucket, replacing any existing roles on the instance.



5. Kinesis Data Firehose Delivery Stream Setup

Within the AWS console, a Kinesis Data Firehose delivery stream was configured with the following options:

- Source: Amazon Kinesis Data Streams (the previously created "logstream")
- Destination: Amazon S3 with the bucket "s3://logbucket-198429" and the prefix format "s3://bucket"

- Buffer hints: 5 MiB buffer size and 60 seconds buffer interval (with recommended compression and file extension settings)

Choose source and destination

Specify the source and the destination for your Firehose

Source | [Info](#)

Amazon Kinesis Data Streams

Destination | [Info](#)

Amazon S3

Destination settings [Info](#)

Specify the destination settings for your Firehose stream.

S3 bucket

s3://logbucket-198429

Format: s3://bucket

New line delimiter

6. AWS Kinesis Agent Installation on EC2

The AWS Kinesis Agent was installed on the EC2 instance (Amazon Linux) using the command `sudo yum install -y aws-kinesis-agent`. The installation completed successfully.

```
~]$ sudo yum install -y aws-kinesis-agent
```

7. Application Directory and Log File Creation

A directory `/opt/myapp` was created with `sudo mkdir /opt/myapp`. Ownership was changed recursively to `ec2-user` to allow write access (`sudo chown ec2-user:ec2-user -R /opt/myapp`). An empty log file `/opt/myapp/myapp.log` was created.

```
complete!  
[ec2-user@ip-172-31-10-10 ~]$ sudo mkdir /opt/myapp  
[ec2-user@ip-172-31-10-10 ~]$ sudo mkdir /opt/myapp  
[ec2-user@ip-172-31-10-10 ~]$ sudo chown ec2-user:ec2-user -R /opt/myapp  
[ec2-user@ip-172-31-10-10 ~]$
```

8. Kinesis Agent Configuration

A configuration file, `agent.json`, was downloaded from a CloudFront distribution and moved to `/etc/aws-kinesis/agent.json` (overwriting if necessary). This configuration defines the monitored log file, destination stream, and parsing rules for the agent.

```
[ec2-user@ip-172-31-10-10 ~]$ sudo mv -f agent.json /etc/aws-kinesis/agent.json  
[ec2-user@ip-172-31-10-10 ~]$
```

9. Starting the Kinesis Agent Service

The Kinesis Agent service was started on the EC2 instance via `sudo service aws-kinesis-agent start`. The service successfully loaded and began running.

```
sudo mv -f agent.json /etc/aws-kinesis/agent.json  
sudo service aws-kinesis-agent start  
[ OK ]  
systemctl status aws-kinesis-agent  
[ OK ]
```

10. Log Entry Generation

Sample log entries were appended to `/opt/myapp/myapp.log` to simulate application logging. This process included writing several lines to the log file and verifying the contents using `cat /opt/myapp/myapp.log`.

```

s sent successfully to destinations. Uptime: 750840ms
[ec2-user@ip-172-31-██████████ ~]$ cat /opt/myapp/myapp.log
This is the first log entry
Line 2 in the log
Line 3 in the log
Line 4 in the log
Line 5 in the log
[ec2-user@ip-172-31-██████████ ~]$

```

11. Kinesis Agent Log Monitoring

The agent's log file was monitored with `tail -25 /var/log/aws-kinesis-agent/aws-kinesis-agent.log`. The output indicated that the agent was reading from the configuration file, found one configured flow, was using up to twelve threads for sending data, and had successfully parsed and sent one record (28 bytes) to the destination. Uptime and progress metrics were also displayed.

```

ully sent 1 records to destination.
2025-12-27 22:58:57.726+0000 (Agent.MetricsEmitter RUNNING) com.amazon.kinesis.streaming.agent.Agent [INFO] Agent: Progress: 1 records parsed (28 bytes), a
s sent successfully to destinations. Uptime: 690839ms

```

12. S3 Data Verification

Validation was performed in the S3 console, where the `logbucket-198429` bucket displayed a new `2025/` folder, likely created based on the year prefix in Firehose delivery. This confirmed that logs were successfully streamed and stored in the bucket.

☐	Name	☐	Type	☐	Last modified	☐	Size	☐	Storage class
☐	2025/	☐	Folder	☐	-	☐	-	☐	-

Project Summary

In this project, the AWS Kinesis Agent was deployed on an EC2 instance to automate the collection and streaming of application logs to Amazon S3. The agent was configured using a custom `agent.json` file that specified the log file to monitor and the target Kinesis Firehose stream. After successfully starting the agent service, test log entries were generated to ensure proper operation. Monitoring the agent's own logs confirmed successful parsing and transmission of data. Finally, validation in the S3 console verified that log files were ingested and stored as expected, demonstrating a complete end-to-end pipeline for reliable log delivery from EC2 to S3.

Architecture:

